

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I **Atmakuri Sai Mrudula/192372087** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Sai Mrudula

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

1. Explain the difference between HDFS and traditional file systems.

HDFS (Hadoop Distributed File System) is designed to store and process very large amounts of data across multiple computers in a cluster. It divides large files into blocks and distributes them across different nodes. It also provides replication and fault tolerance. Traditional file systems generally store files on a single computer or server and are not designed for large-scale distributed data processing.

2. Why is replication important in distributed storage systems?

Replication is important because it keeps multiple copies of data on different nodes. If one node fails, another copy can be accessed without losing the data. It improves data availability, reliability, and fault tolerance. In Hadoop, HDFS commonly uses a replication factor of three to protect data from hardware failures.

3. How does MapReduce divide work between map and reduce phases?

MapReduce divides a large processing task into Map and Reduce phases. In the Map phase, input data is divided into smaller parts and processed in parallel to generate key-value pairs. During the Reduce phase, values having the same key are grouped and processed to produce the final output. This approach allows Hadoop to process large datasets efficiently.

4. What role does YARN play in large-scale Hadoop processing?

YARN stands for Yet Another Resource Negotiator and is responsible for managing resources and scheduling applications in a Hadoop cluster. It allocates CPU and memory resources to different jobs and monitors their execution. YARN allows multiple applications to share the same Hadoop cluster efficiently and improves overall resource utilization.

5. Differentiate NAS and SAN in an enterprise data environment.

NAS (Network Attached Storage) provides file-level storage over a network and is mainly used for file sharing, backups, and general storage. SAN (Storage Area Network) provides block-level storage and is commonly used for high-performance databases and enterprise applications. NAS is generally simpler and less expensive, while SAN provides higher performance and more advanced storage capabilities.

6. What is the purpose of ETL in a big data pipeline?

ETL stands for Extract, Transform, and Load. It extracts data from different sources, transforms it by cleaning, filtering, or converting it into a suitable format, and finally loads it into a data warehouse, data lake, or other storage system. ETL helps convert raw data into clean and useful information for analysis and decision-making.

7. How does Apache NiFi help in data integration workflows?

Apache NiFi is a data integration and flow-management tool used to move and process data between different systems. It can collect data from databases, APIs, applications, and log files and transfer it to systems such as Hadoop or cloud storage. NiFi also provides data routing, transformation, monitoring, and tracking of data flow.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Common Hadoop failures include DataNode, NameNode, disk, network, and ResourceManager failures. Hadoop handles these failures using data replication, NameNode and ResourceManager high availability, monitoring, and automatic recovery. For example, if a DataNode fails, HDFS can use another replicated copy of the data and create a new replica when necessary.

9. How would you design a secure data ingestion pipeline for a large organization?

A secure data ingestion pipeline should use authentication, authorization, encryption, data validation, and monitoring. Data should be encrypted during transmission using secure protocols and encrypted while stored. Role-based access control, audit logs, network security, and sensitive-data masking should also be implemented. This provides end-to-end protection from the data source to the storage and analytics layers.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

A Hadoop system can be explained as a group of many computers working together to handle a large amount of data. Instead of depending on one powerful computer, Hadoop distributes data and processing across multiple machines. When the amount of data increases, more machines can be added to the cluster. This makes Hadoop scalable, fault tolerant, and suitable for processing very large datasets.